

Lumen electric power steering torque sensor fault causing intermittent assist

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Area: bulletins

1 Condition

Affected Lumen vehicles may experience intermittent loss of electric power steering assist accompanied by DTC C1556 (EPS Torque Sensor Fault) and illumination of the EPS warning lamp. The condition is most often felt as a momentary increase in steering effort followed by restoration of assist as the torque sensor signal recovers. The fault is associated with the electric power steering assembly EPS-8740. For an overview of the assist motor and integrated sensor, see Lumen Steering Service Manual :: Electric Power Steering Motor .

2 Cause

The cause is an intermittent signal dropout from the torque sensor internal to electric power steering assembly EPS-8740, which sets C1556 when the redundant sensor channels disagree. Thermal expansion of an internal solder joint produces the intermittent nature of the fault, so the condition may not be present during a static inspection. Replacement of the EPS-8740 assembly is required, as the torque sensor is not separately serviceable. Code-setting criteria are detailed in Lumen Steering Service Manual :: EPS Torque Sensor Fault .

3 Repair Procedure

Diagnose the concern using PROC-EPS-DIAG, capturing C1556 freeze-frame data and monitoring both torque sensor channels during a wiggle and warm-up test before condemning the assembly. The full diagnostic flow is documented in the Lumen Steering Service Manual :: Electric Power Steering Diagnosis Procedure ; see also the Lumen Steering Service Manual :: Diagnostic Trouble Codes for related codes. After replacing electric power steering assembly EPS-8740, perform a torque sensor calibration and a four-wheel alignment per PROC-ALIGN as described in the Lumen Steering Service Manual :: Four-Wheel Alignment Procedure . Clear C1556, road test, and confirm assist no longer drops out.